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Examining the Hokuriku Shinkansen's Tsuruga–Shin-Osaka Extension Against Its Own Record of Wrong Forecasts and Left-Out Costs

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1.0 Introduction

14 In June 2026 the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) published four benefit-cost ratios for the same 139 kilometres of railway. Evaluated across the whole Tokyo–Shin-Osaka corridor, the Obama-Kyoto route returns a ratio of 1.1 at a 4% social discount rate and 1.3 at 2%. Evaluated on the unbuilt Tsuruga–Shin-Osaka segment alone, the same route returns 0.5 and 0.9 [1], [2]. The railway is identical across all four figures, yet two choices differ. Firstly, where the boundaries are decided, which MLIT widened for the first time to the whole corridor in 2026 [1], and finally which discount rate makes the headlines. The resulting range straddles 1.0 on the threshold for construction.

16 Japan's Shinkansen lines from the 1973 plan are built and owned by the Japan Railway Construction, Transportation and Technology Agency (JRTT) and leased to the JR operators. After the lease is paid by the operating company, construction is split 2/3rds nationally and 1/3rd for the prefecture. Appraisals follow a MLIT manual, which sets 4% as a primary discount rate, allows a lower reference rate and describes a 30 or 50 year evaluation period [3]. A ratio of 1.0 functions as the conventional benchmark against which appraisals are reported and contested, but is one of five conditions required for construction to start. In July 2026 the Katsuragawa variant was adopted at 139 km with roughly 26 years of construction and planned opening of 2054, yet has estimated costs that rose from ¥2.1 trillion in 2016 to ¥3.4 trillion by 2024 [4], [5].

Three problems compound in key decision points. First, the reliability of the programme's own demand forecasts has been questioned, including within the ex-post evaluations that JRTT publishes on its completed sections [6]. Second, the horizon is extreme. The forecast runs from 28 to 50 years to opening, however Japan's population is forecasted to shrink by 105 million people by 2050 [7]. Third, the appraisal reaches the public as a small set of point ratios measured against a fixed benchmark. The mismatch is between the confidence of that output and the uncertainty of its inputs.

Closing that gap requires three constructs to be defined rather than assumed, because each looks like a measurement and is in fact a choice. Forecast error is not simply forecast minus actual: it conceals appraisal, operator and promotional methods. Hence, only displayed errors are corrected. Furthermore, the Benefit-cost ratio is not simply benefits over costs: it measures the project, evaluation perimeter, the discount rate, the valuation of travel time and the appraisal period, as the four opening figures demonstrate. In the appraisals, local costs are omitted from the with/without comparisons and thus are counted separately and therefore no Government tier owns and mitigates the risk this poses to forecasting error.

The Research Question asks: To what extent does the Tsuruga to Shin-Osaka extension's benefit-cost ratio remain above 1.0 when demand and cost inputs are corrected using the Seibi-Shinkansen project's forecasting records (1997-2016) and local costs outside appraisal are added?

The study compiles signed ex-ante versus ex-post ridership errors for Seibi Shinkansen sections opened between 1997 and 2024, establishes whether that error distribution legitimately transfers

to the current appraisal, and applies it through a sensitivity grid to test whether the extension's ratio survives its own corridor's history.

2.0 Existing Literature

The foundational finding is Flyvbjerg, Holm and Buhl's study of 210 transport infrastructure projects, which reports rail passenger demand overestimated by an average of 106% with no improvement over the thirty years sampled [8]. Its contribution here is not the magnitude but the character of the error: forecast inaccuracy is systematic rather than random, which is the precondition for the need to correct it. Flyvbjerg subsequently formalised that correction as reference class forecasting, in which an official estimate is adjusted using the outcome distribution of comparable completed projects; its first application, to the Edinburgh Tram with the UK Department for Transport, gives the method for the Government rather than purely academic standing [9]. A recent review documents where the method fails, principally in class selection and in the instability of small samples [10]. Alongside this, Vickerman argues that high-speed rail's transformative claims are contested and that its effects may be redistributive rather than net-additive, which supports treating costs borne locally as real rather than as accounting residue [11].

No clear application of the methodology was seen with the Seibi-Shinkansen line. One appraisal agency was using a standard format and reported forecasted and realised data in the same document. Additionally the directional premise has not been tested where errors can go both ways. On the Shin-Yatsushiro-Kagoshima-Chuo section, year one ridership of 8,800 a day actually beat JR Kyushu’s forecast of 6,800 by 30% [6]. One explanation could be that JR’s lease fee is fixed independently from profit over 30 years, thus incentivising the operator to forecast lower. This is the opposite of what Flyvbjerg identified with approval incentives [8]. This study extends the method to a new jurisdiction and accepts that forecasting errors is systemic but evaluates whether the sign of that error is universal or just jurisdictional.

3.0 Methodology

The study is a quantitative document-based analysis in three stages, corresponding to the three hypotheses set out in §4.

3.1 Reference class and scope.

The reference class comprises Seibi Shinkansen sections only. The Tokaido and Sanyo lines are excluded despite being Shinkansen because they predate the framework and have no comparable ex-ante appraisal; this is a comparability requirement, not a convenience, since reference class forecasting is valid over projects sharing a forecasting process rather than a physical form [9]. Candidate sections, each to be confirmed as having a published evaluation containing both forecast and realised ridership before it is admitted: Takasaki–Nagano (1997), Morioka–Hachinohe (2002), Shin-Yatsushiro–Kagoshima-Chuo (2004), Hachinohe–Shin-Aomori (2010), Hakata–Shin-Yatsushiro (2011), Nagano–Kanazawa (2015), Shin-Aomori–Shin-Hakodate-Hokuto (2016), giving an n of seven. Kanazawa-Tsuruga(2024) awaits evaluation and is reported qualitatively from JR West’s Data [14]). The target case is

Tsuruga–Shin-Osaka, Katsuragawa variant. The unit of analysis is not the project but the error: one signed ex-ante/ex-post ridership pair per section.

3.2 Data sources (all data is public)

Category	Holder	Contribution
Standardised ex-post appraisals	JRTT project evaluation archive (事業評価) [6]	Forecast and realised ridership, B/C and EIRR in one format across sections. The core dataset
Audit target	MLIT Railway Bureau / JRTT explainer, Oct. 2024 [4]	The official point forecast, methodology statement and lease-fee ruling
Current appraisal outputs	MLIT recalculation as reported June 2026 [1], [2]	The 0.5 / 0.9 / 1.1 / 1.3 starting values
Cost history	MLIT cost revision, Kanazawa–Tsuruga, 2021 [13] and 2024 explainer [4]	18.8% Kanazawa-Tsuruga overrun, estimation rising from ¥2.4 to ¥3.4 trillion
Operator actuals	JR West ten-year usage release, 2025 [14]	Independent realised ridership for triangulation
Calculation rules	MLIT railway project evaluation manual [3]	Admissible benefit categories, discount rate, appraisal period

Development Bank of Japan forecast and outcome studies for Ishikawa [12], [15] are excluded as they measure impact in Yen, not ridership numbers. .

3.3 Source evaluation

Each category is assessed against CRAAP and OPCVL criteria, but the structural problem is sharper than either framework captures. JRTT evaluates its own projects, so forecasters and auditors are the same institution. This is stated once and used to justify triangulating official realised figures against JR West operator data. Government forecasts additionally carry a documented directional tendency [8], so official projections are treated as interested evidence rather than as neutral measurement. Sources are Japanese-language; extraction consists of reading tabulated figures rather than interpretive translation, which keeps the feasibility claim honest.

3.4 Design and procedure

The design is paired ex-ante versus ex-post within each case, pooled across cases into a distribution. The pairing is what generates an error term at all: realised ridership alone describes

demand, whereas the gap describes the institution's forecasting behaviour. It is also the only way to get to a target case that cannot be observed until 2054.

Stage one compiles and works out signed errors, where e=(realised-forecast)/forecast. Here the mean, median sign pattern and range are defined.

Stage two splits errors with the 2010 demand-method revision which was recorded in JRTT's evaluation [6], and compares approval-stage forecasts made before with re-evaluation forecasts made after the fact. This tests whether the revision changed the size or sign of errors and can be checked with current forecasts to define whether current forecasts used the revised method.

Stage three constructs a sensitivity grid: demand percentiles (P10, P25, P50, P75) drawn from that distribution, crossed with cost scenarios (official estimate, the +18.8% corridor overrun rate, and the reference-class mean overrun from initial to final costs [6]) , reported separately at both perimeters and both discount rates. Each cell applies adjusted benefits/costs defined by the formula below where o is cost-overrun and L is excluded local resource costs, mainly prefectural support for the operators of separate but parallel lines, where [3] confirms this is excluded.

$$B/C = \frac{B(1+e)}{C(1+o)+L}$$

The prefectural construction share and the JRTT lease [16] are transfers already inside C (costs) and therefore are reported in a separate table showing which stakeholder pays, with a possible revision of the 2:1 ratio[5] as a scenario that can't be ignored if it occurs, thus acts as a scenario parameter.

3.5 Controls

Three controls are applied and stated. Realised ridership must be matched to the forecast year and unit, passengers per day per km, as each appraisal specifies. This avoids the opening-year surge numbers and keeps forecasting consistent. Forecast lead time is recorded per section, because error scales with horizon and pooling a fifteen-year and a five-year forecast would partly measure how far ahead each agency chose to forecast. Any section whose measurement window includes 2020–22 is flagged for COVID distortion.

3.6 Limitations

Seven observations give a wide distribution, a documented weakness of method in [10] and the comparison from 2010 resides on the post-revision forecasts, which is direction rather than accuracy [10]. Stage three scales a published ratio by adjusted inputs rather than rebuilding MLIT's model, because the full benefit stream is not disaggregated publicly; this assumes benefits scale approximately linearly with passenger volume, and the assumption is stated as a bound on the result. Under-forecasting may indicate shocks not the method, the Nagano-Kanazawa line opened in 2015 during a high influx of tourism, although the target case now will open into a Japan with a shrinking population by 2054. Finally, the method yields ranges rather than point predictions, and it cannot establish whether the extension was the most effective use of capital, only whether it satisfies the criterion by which it is publicly defended.

4.0 Hypotheses

H1. Official pre-construction ridership forecasts for the seven completed sections deviate from realised outcomes by more than 20% in absolute terms. This alone is enough to carry the 2% segment ratio of 0.9 to 1, and are on aggregate positive, meaning realised comes in above forecast.

H2. The forecasting method has been revised in specification (2010), but no correction derived from the programme's own realised error is publicly documented, the error was reduced rather than removed. Historical error distribution stands as a lower-bound prior on the Tsuruga-Shin-Osaka forecast, and this project’s 28 year lead time differs greatly from the reference class’ lead time.

H3 (stage 3 MLIT’s position). Evaluated on the integrated Tokyo to Shin-Osaka perimeter at a 4% social discount rate, the extension's benefit-cost ratio remains at or above 1.0 once demand and cost scenarios are corrected by that record, and adding local resource costs.

Flyvbjerg establishes that forecast error in public works is systematic rather than random [8]. The direction of that error is predicted rather than assumed, with the one verified pair on this corridor running opposite to this sample [6]. H1 and H2 are expected to be supported and together establish the premise; H3 is MLIT's own claim and is expected to break on the cost axis rather than the demand axis, as the extensions estimate has risen by 62% even before an inflation-adjustment[4].

H3 may run true, and that possibility is stated in advance. If demand errors skew forecasts towards being pessimistic, corrected benefits rise and may offset cost escalation; that result will be reported as found. H3 is specified at the integrated perimeter because the published segment-only ratio at 4% is already 0.5, and a hypothesis that is false before any correction is applied tests nothing and would deliver little insight.

5.0 Significance

Research on forecast accuracy is almost entirely retrospective: it establishes that a project went wrong once the money is spent. Flyvbjerg's case that reference class forecasting is most effective before money is spent, at the point the decision is made [9], and this study sits at that point. Construction start has slipped toward FY2027, funding is unsettled, and the national to local cost split is under revision [5], leaving open how much the prefectures, who fund a third of construction and inherit the parallel line, will bear.

If at 4%, the evaluation perimeter moves the verdict by 0.6 points while demand uncertainty moves it less, approval turns on where a boundary was drawn rather than on what the railway does, and a benchmark met by redrawing that boundary is not a constraint. The reform implied is not "forecast better" but fixing the perimeter by rule: incremental costs of the segment under decision against marginal benefits wherever they fall. For the existing literature, under-forecasting would imply that the sign of forecasting error is institutional, not universal.

References

[1] "Hokuriku Shinkansen: Obama–Kyoto route shows the highest cost-effectiveness; MLIT estimates for Tokyo–Shin-Osaka and the seven other proposals," Fukui Shimbun, Jun. 11, 2026. Accessed: Sept. 19, 2026. [Online]. Available: https://www.fukuishimbun.co.jp/articles/-/2623667 (in Japanese).

[2] "Hokuriku Shinkansen extension: Obama–Kyoto route superior on cost-effectiveness in new MLIT estimates," Nihon Keizai Shimbun, Jun. 11, 2026. Accessed: Sept. 19, 2026. [Online]. Available: https://www.nikkei.com/article/DGXZQOUA112ZQ0R10C26A6000000/ (in Japanese).

[3] Ministry of Land, Infrastructure, Transport and Tourism, Railway Bureau, "Manual of evaluation methods for railway projects, 2012 ed. (partially revised Mar. 2025)," Tokyo, Japan, Jul. 2012. Accessed: Sept. 19, 2026. [Online]. Available: https://www.mlit.go.jp/common/000224631.pdf (in Japanese).

[4] Ministry of Land, Infrastructure, Transport and Tourism, Railway Bureau, and Japan Railway Construction, Transport and Technology Agency, "Explanatory materials on the Hokuriku Shinkansen (Tsuruga–Shin-Osaka section)," Tokyo, Japan, Oct. 2024. Accessed: Sept. 19, 2026. [Online]. Available: https://www.jrtt.go.jp/project/turuhannrennrakukaigi08.pdf (in Japanese).

[5] "LDP and JIP decide on Kyoto route for Hokuriku Shinkansen line extension," The Japan Times, Jul. 15, 2026. Accessed: Sept. 19, 2026. [Online]. Available: https://www.japantimes.co.jp/news/2026/07/15/japan/society/hokuriku-shinkansen-katsuragawa/

[6] Japan Railway Construction, Transport and Technology Agency, "Project evaluation archive." Accessed: Sept. 19, 2026. [Online]. Available: https://www.jrtt.go.jp/construction/committee/ (in Japanese).

[7] National Institute of Population and Social Security Research, "Population projections for Japan, 2023 revision," Tokyo, Japan, Apr. 2023. Accessed: Sept. 19, 2026. [Online]. Available: https://www.ipss.go.jp/pp-zenkoku/j/zenkoku2023/pp_zenkoku2023.asp (in Japanese).

[8] B. Flyvbjerg, M. K. Skamris Holm, and S. L. Buhl, "How (in)accurate are demand forecasts in public works projects? The case of transportation," J. Amer. Planning Assoc., vol. 71, no. 2, pp. 131–146, 2005, doi: 10.1080/01944360508976688.

[9] B. Flyvbjerg, "Curbing optimism bias and strategic misrepresentation in planning: Reference class forecasting in practice," Eur. Planning Stud., vol. 16, no. 1, pp. 3–21, 2008, doi: 10.1080/09654310701747936.

[10] C. C. Cantarelli, K. Davis, J. K. Pinto, and N. Turner, "Reference class forecasting: Promises, problems, and a research agenda moving forward," Prod. Planning Control, vol. 37, no. 7, pp. 691–709, 2026, doi: 10.1080/09537287.2025.2578708.

[11] R. Vickerman, "Can high-speed rail have a transformative effect on the economy?" Transp. Policy, vol. 62, pp. 31–37, 2018, doi: 10.1016/j.tranpol.2017.03.008.

[12] West Japan Railway Company, "Hokuriku Shinkansen usage over ten years," Osaka, Japan, Mar. 17, 2025. Accessed: Sept. 19, 2026. [Online]. Available: https://www.westjr.co.jp/press/article/items/250317_00_press_hokurikushinkansen_goriyoujyoukyou.pdf (in Japanese).

[13] Ministry of Land, Infrastructure, Transport and Tourism, "Revision of construction costs, Hokuriku Shinkansen Kanazawa–Tsuruga section," Tokyo, Japan, 2021. Accessed: Sept. 19, 2026. [Online]. Available: https://www.mlit.go.jp/report/press/content/001397617.pdf (in Japanese).

[14] Development Bank of Japan, Hokuriku Branch, "First-year effects of the Hokuriku Shinkansen opening," Kanazawa, Japan, Dec. 2016. Accessed: Sept. 19, 2026. [Online]. Available: https://www.dbj.jp/pdf/investigate/area/hokuriku/pdf_all/hokuriku_1612_02.pdf (in Japanese).

[15] Development Bank of Japan, Hokuriku Branch, "Economic effects of the Hokuriku Shinkansen opening," Kanazawa, Japan, Mar. 2013. Accessed: Sept. 19, 2026. [Online]. Available: https://www.dbj.jp/upload/investigate/docs/hokuriku_1303_01.pdf (in Japanese).

[16] Ministry of Land, Infrastructure, Transport and Tourism, "Authorisation of lease charges," Tokyo, Japan, 2024. Accessed: Sept. 19, 2026. [Online]. Available: https://www.mlit.go.jp/report/press/tetsudo03_hh_000109.html (in Japanese).